

Case 1: EURIBOR

1. Summary

The objective was to identify a bounded-risk strategy that capitalizes on a "Stability" view while maximizing the width of the profit zone. The analysis recommends the **Put Condor centered at 98.05**.

- **The Winner:** Put Condor **98.05** (Wing Width = 0.15).
- **Dominance:** This structure offers the highest **Expected Value (11.28)** relative to the width of the profit zone. While slightly more expensive than deeper OTM structures, it aligns perfectly with the center of the client's target zone (98.00–98.12), maximizing the probability of capture.
- **Structural Alpha:** The strategy monetizes the "Crash Skew." By selling the downside body strikes (which are trading at a panic premium), the Condor subsidizes the cost of the wider coverage.

2. The Model: Optimization & Calibration

Modeling this market required solving for a severe dislocation: a "Cheap" ATM volatility (0.75 ticks) co-existing with an "Explosive" Downside volatility (4.50 ticks).

Challenge 1: The "Broken" Volatility Surface

The Problem: Standard calibration algorithms effectively "flat-lined," failing to reconcile the steep drop in ATM prices with the massive premium in the wings.

The Solution: I implemented a Weighted Least-Squares Optimization.

Instead of manually tuning parameters (as I had first done after multiple failed calibration attempts), I configured a solver to minimize the error between the model price and the market benchmarks, with a heavy penalty weighting on the ATM anchor price. This forced the model to respect the liquidity of the center strike while stretching to fit the wings.

The optimizer converged on a specific set of SABR Normal parameters that describe this dislocation:

- **vol of vol = 2.19:** This extreme value (standard is ~0.30) confirms the "Explosive Wings" hypothesis. It forces the model to create massive convexity, explaining why the downside fly trades at 4.50 ticks.
- **skew = 0.91:** A near-perfect negative correlation captures the "Crash Tilt". It directs the volatility explosion almost entirely to the downside (Puts).
- **alpha = 0.48:** To support such extreme convexity without collapsing the ATM price to zero, the model requires a higher base volatility.

Challenge 2: Model Selection

Decision: We utilized Bachelier (Normal) pricing rather than Black (Log-normal).

Reasoning: In STIR markets, log-normal models distort skew analysis when rates are low or negative. Bachelier treats volatility in absolute basis points, providing a stable ruler to measure the "Wide Zone" probabilities.

3. Strategy Framework: The "Trader View"

To select the winning trade, we compared two distinct probability distributions:

Model A: The "Box View" (Accepted)

- **Assumption:** The client's view is "Stability" somewhere between 98.00 and 98.12.
- **Distribution: Uniform (Box).**

This mathematically rewards Width. It tells the model to focus on getting in the zone rather than a single peak. This framework correctly identifies the Condor as the winner because its flat-topped payoff aligns with the flat probability distribution.

Model B: The "Gaussian View" (Rejected)

- **Assumption:** Standard Bell Curve distribution.
- **Impact:** This concentrates probability at the exact peak (98.06). It biases the results toward narrow Flies, which have high peaks but narrow safety nets. We rejected this because it fails to capture the client's preference for a "wide profit zone".

4. Final Analysis: Why the Condor Wins

The recommendation is the **Put Condor 98.05**.

Cost vs. Value

A Condor centered at 97.97 is cheaper (Cost ~0.78) because it sells deeper into the skew, but it misaligns with the client's target, missing the upper half of the zone. The Condor 98.05 costs more (~1.40 ticks) but generates a significantly higher Expected Value (EV 11.28). Rationally, you pay the extra premium to align the structure's "sweet spot" exactly with the highest probability outcome (98.06).

Skew Monetization

The structural advantage of the Condor comes from its wider wings. Even centered at 98.05, the Condor sells strikes (e.g., 98.03) that are beginning to price in the skew premium. This generates a higher credit per unit of risk compared to a standard Butterfly, effectively subsidizing the cost of the wider body.

Risk Note: Negative Carry

The trade carries Negative Carry (-0.96 ticks/mo). Being short delta (targeting a zone below Spot 98.11) means we are fighting the yield curve slide. This cost is unavoidable for a downside target, but the "Box View" optimization ensures we maximize the probability of the future sliding into our trap, making the carry cost acceptable.

Source code in .py file in email